POULTRY PRODUCTION COST OPTIMIZATION

& FINANCIAL FEASIBILITY

A real-world operations analysis using R, scenario modelling and business decision frameworks

PORTFOLIO CASE STUDY

Broiler production operation | Anambra State, Nigeria
Historical cycle: approximately November–December 2025

BY

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Executive finding: The recorded cycle generated approximately ₦830,000 in sales revenue against ₦305,900 in documented/approximate costs, giving a preliminary surplus of ₦524,100 before other unrecorded expenses. The analysis identifies a potential feed-cost opportunity but treats in-house feed savings and larger-scale outcomes as hypotheses to test, not proven results.

Prepared as a reproducible analytics portfolio project

1. Executive summary

This project converts an actual broiler production cycle into a transparent business analysis and a structured decision model. It investigates production performance, commercial-feed cost, a proposed in-house feed alternative, and a sequential expansion strategy from 1,000 to 3,000 and then 5,000 birds. The longer-term business concept includes hatching fertilized eggs and investing in a feed mill, solar power, a truck and an incubator.

> The key analytical discipline is separating observed results from estimates. The 50-bird cycle is the historical baseline. The in-house feed alternative was not implemented in that cycle. The 1,000/3,000/5,000-bird calculations are illustrative linear scenarios, not a forecast or investment recommendation.

Business question

How can the farm improve unit economics and expand while retaining enough working capital for the next cycle, maintaining a prudent cash reserve, and determining how much excess cash may be withdrawn for other business purposes?

2. Historical performance

Metric	Value	Interpretation
Birds placed	50	Actual
Birds sold	48	Actual
Birds died	2	Actual
Survival rate	96.0%	48 sold / 50 placed
Mortality rate	4.0%	2 died / 50 placed
Sales revenue	₦830,000	Approximate actual total revenue
Documented/approximate costs	₦305,900	Chicks + feed + medication/supplements
Preliminary surplus	₦524,100	Before other unrecorded expenses
Average revenue per bird sold	₦17,291.67	Revenue / 48 sold
Known cost per bird sold	₦6,372.92	Known costs / 48 sold

Cost breakdown

Cost item	Amount	Evidence status
Day-old chicks	₦102,500	Documented actual
Commercial starter feed	₦100,400	Documented actual
Commercial finisher feed	₦86,000	Documented actual
Medication and supplements	≈ ₦17,000	Approximate actual
Total	₦305,900	Known/documented total

> The preliminary surplus is not confirmed net profit. Labour, litter, water, transport, fuel, cleaning, and any veterinary expenses outside the reported amount have not been quantified.

3. Feed make-or-buy analysis

The documented commercial-feed cost was ₦186,400 (₦100,400 starter + ₦86,000 finisher). The original research estimated in-house ingredient costs of ₦90,000 for the same 50-bird batch (₦45,000 starter + ₦45,000 finisher).

> The simple arithmetic difference is ₦96,400, equal to 51.72% of the commercial-feed amount. This is a potential cost gap to investigate, not an observed saving.

Costs omitted from the in-house estimate

The estimate did not include transport of ingredients, milling, fuel/power, labour, wastage, feed formulation and quality control, equipment maintenance, depreciation, or the capital cost of the truck, mill and solar system. Buying a truck or solar system does not make the associated operating costs zero. Feed quantities and formulas must be validated for performance and safety before commercial use.

Business implication

The next analysis should calculate a fully loaded in-house cost per kilogram and per bird, then compare it with commercial feed using equivalent nutritional specifications. A small pilot batch and feed-quality/performance monitoring would reduce the risk of scaling an untested estimate.

4. Sequential scale-up model

The intended expansion is sequential: produce and sell 1,000 birds, then proceed to 3,000, then 5,000. Profits or excess cash may be withdrawn for other business purposes or retained as poultry savings, but working capital for the next cycle must be protected first.

For demonstration, the model scales the historical revenue and known costs linearly from the 50-bird baseline. It assumes the same 96% survival rate, same average revenue per bird sold and same unit costs. This makes the calculations reproducible, but does not capture economies of scale, bulk purchasing, new labour/facility costs, feed wastage, market demand or capital expenditure.

Target placed	Illustrative birds sold	Illustrative revenue	Known costs	Preliminary surplus*
1,000	960	₦16,600,000	₦6,118,000	₦10,482,000
3,000	2,880	₦49,800,000	₦18,354,000	₦31,446,000
5,000	4,800	₦83,000,000	₦30,590,000	₦52,410,000

*Before other operating costs, capital expenditure, finance costs and any omitted expenses. Figures are linear illustrations only—not forecasts.

5. Sensitivity and risk

The sensitivity table tests how the illustrative 5,000-bird result changes when sale price falls, survival declines, or known unit costs rise. The scenarios do not predict that these events will occur; they demonstrate which risks should be quantified.

Scenario	Survival	Revenue (illustrative)	Known costs only	Preliminary surplus*
Baseline-equivalent illustration	96%	₦83,000,000	₦30,590,000	₦52,410,000
Selling price down 10%	96%	₦74,700,000	₦30,590,000	₦44,110,000
Known unit costs up 10%	96%	₦83,000,000	₦33,649,000	₦49,351,000
Survival falls to 92%	92%	₦79,541,667	₦30,590,000	₦48,951,667
Price down 10% and known costs up 10%	96%	₦74,700,000	₦33,649,000	₦41,051,000
Price down 10%, survival 92%, costs up 10%	92%	₦71,587,500	₦33,649,000	₦37,938,500

*All sensitivity results exclude unrecorded expenses and capital expenditure. Do not use them as a financing decision until the missing inputs are populated.

6. Cash allocation framework

At the end of each cycle, the farm should first calculate cash after all cycle costs and obligations. It should then reserve the required working capital for the next cycle. Only the balance after the next-cycle reserve and any chosen emergency reserve is available for withdrawal.

Maximum safe withdrawal = cash available − next-cycle working capital − outstanding obligations − chosen reserve.

The conservative strategy should compare reserve targets equal to one, two and three months of operating expenses. The exact withdrawal amount is intentionally not calculated in this version because actual next-cycle operating costs, asset quotes and other obligations are not yet available.

7. Hatchery and vertical integration

The planned hatchery model should compare the fully loaded cost per usable chick from fertilized eggs with the cost of buying day-old chicks. It requires egg price, delivery, fertility, hatchability, usable-chick rate, incubator capital cost, power, labour, sanitation, maintenance and early chick mortality. None of these unknowns should be replaced with invented farm results.

8. Recommended BI dashboard

Dashboard page	Key visuals / measures	Decision supported
Executive overview	Revenue, known costs, preliminary surplus, survival and mortality KPIs	Understand historical baseline
Cost breakdown	Chick, starter feed, finisher feed, medication; missing-cost flag	See cost concentration and evidence gaps
Feed make-or-buy	Commercial feed cost vs estimated in-house cost; omitted-cost checklist	Decide whether to pilot in-house feed
Scale-up scenarios	1k/3k/5k stage comparison; visible illustrative label	Compare scale mechanics without overstating certainty
Cash allocation	Next-cycle working capital, reserve target, maximum safe withdrawal	Protect liquidity while funding other businesses
Sensitivity	Sale price, survival and unit-cost shock scenarios	Identify drivers of downside risk

9. Conclusions and next actions

The historical cycle shows 96% survival and approximately ₦830,000 in sales revenue. Recorded costs total ₦305,900, producing a preliminary surplus of ₦524,100 before unrecorded expenses. The proposed in-house feed estimate is materially lower than the historical commercial-feed cost, but it has not been tested and excludes important costs. The project therefore supports a structured pilot and better cost recording—not an immediate claim that vertical integration will deliver the estimated savings.

Portfolio integrity note: All reported **historical** figures are based on real farm information and the original project document. Approximate figures remain labelled approximate; unobserved costs and untested strategies are not represented as facts.